

Constructing Triangles, Quadrilaterals, and Regular Polygons

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Triangles are polygons with three sides.
The sum of their interior angles is always 180° .

Quadrilaterals are polygons with four sides.
The sum of their interior angles is always 360° .

A **regular polygon** has all sides equal and all interior angles equal.

When drawing a figure with given angle measures, first sketch the shape, then measure and construct each angle carefully.

Regular Polygons

Common regular polygons include:

Regular pentagon: 5 sides, 108°

Regular hexagon: 6 sides, 120°

Regular octagon: 8 sides, 135°

Regular decagon: 10 sides, 144°

Examples

Example 1

Draw a triangle with angles measuring 50° , 60° , and 70° .

Label all vertices and indicate the given angle measures.

Example 2

Draw a quadrilateral with interior angles measuring

80° , 90° , 100° , 90° .

Label all vertices and indicate the given angle measures.

Example 3

Draw a regular pentagon.

Label all vertices and indicate the angle measures.

Example 4

Draw a regular octagon.

Label all vertices and indicate the angle measures.

Practice Exercises

Practice 1

Draw a triangle with angles measuring 50° , 60° , and 70° .

Label all vertices and indicate the given angle measures.

Practice 2

Draw a triangle with angles measuring 40° , 65° , and 75° .

Label all vertices and indicate the given angle measures.

Practice 3

Draw a quadrilateral with interior angles measuring

80° , 90° , 100° , 90° .

Practice 4

Draw a quadrilateral with interior angles measuring

70° , 110° , 95° , 85° .

Practice 5

Draw a regular pentagon.

Label all vertices and indicate the angle measures.

Practice 6

Draw a regular hexagon.

Label all vertices and indicate the angle measures.

Practice 7

Draw a regular octagon.

Label all vertices and indicate the angle measures.

Practice 8

Draw a regular decagon.

Label all vertices and indicate the angle measures.

Practice 9

Draw a triangle with angles measuring 35° , 55° , and 90° .

Label all vertices and indicate the given angle measures.

Practice 10

Draw a quadrilateral with interior angles measuring

120° , 80° , 70° , 90° .

Activities

Activity 1

Draw a triangle with angles measuring 45° , 55° , and 80° .

Label all vertices and indicate the given angle measures.

Activity 2

Draw a triangle with angles measuring 30° , 70° , and 80° .

Label all vertices and indicate the given angle measures.

Activity 3

Draw a quadrilateral with interior angles measuring

85° , 95° , 105° , 75° .

Activity 4

Draw a quadrilateral with interior angles measuring

65° , 115° , 90° , 90° .

Activity 5

Draw a regular pentagon.

Label all vertices and indicate the angle measures.

Activity 6

Draw a regular hexagon.

Label all vertices and indicate the angle measures.

Activity 7

Draw a regular octagon.

Label all vertices and indicate the angle measures.

Activity 8

Draw a regular decagon.

Label all vertices and indicate the angle measures.

Activity 9

Draw a triangle with angles measuring 25° , 65° , and 90° .

Label all vertices and indicate the given angle measures.

Activity 10

Draw a quadrilateral with interior angles measuring

130° , 70° , 85° , 75° .

Thank you!